DSFB Computer Graphics Reviewer Bundle

This PDF is generated from a single timestamped run directory so reviewers can archive the bounded synthetic artifact without depending on the notebook session state.

"The experiment is intended to demonstrate behavioral differences rather than establish optimal performance."

What This Bundle Contains

Demo A figures, key numeric metrics, an interpretation grounded in the actual run outputs, and the main limitations of the artifact.

The bundle is intended to support technical review, transition conversations, and reproducible replay rather than optimality claims or deployment claims.

Run directory: output-dsfb-computer-graphics-20260323-021943

Key Metrics

On the canonical scenario, host-realistic DSFB reduced cumulative ROI MAE from 2.84366 for fixed alpha to 0.31904.

Against the strong heuristic baseline, host-realistic DSFB changed cumulative ROI MAE from 0.49435 to 0.31904; mixed outcomes are surfaced per scenario below.

Metric	Fixed-alpha	Strong heuristic	Host-realistic
Ghost persistence frames	12	1	0
Peak ROI error	0.43507	0.13843	0.07198
Cumulative ROI error	2.84366	0.49435	0.31904
Average non-ROI MAE	0.01244	0.00062	0.00219

Event Timing

Canonical onset frame: 6

Host-realistic onset response latency: 0

Strong-heuristic onset response latency: 0

Host-realistic trust/error rank correlation: -1.0000

Figure 1. System Diagram

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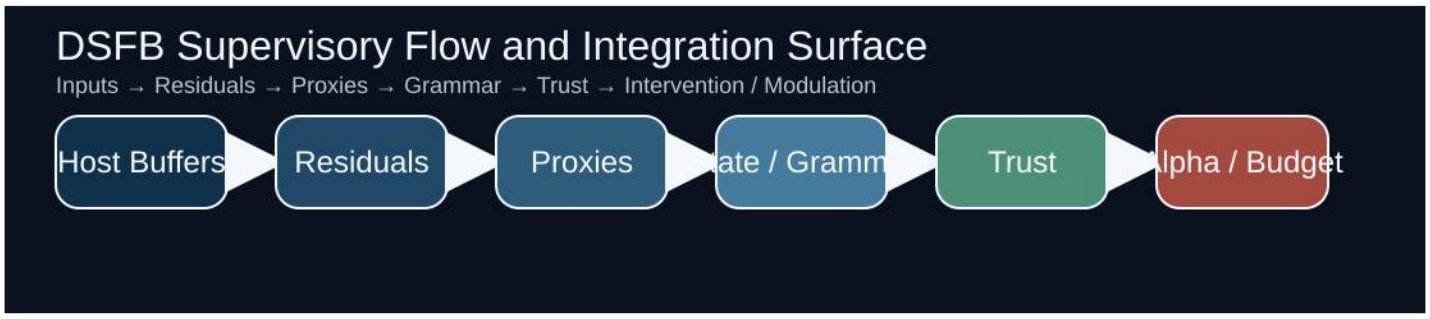

Inputs, residuals, proxies, simplified grammar, trust, and intervention as used in the current crate.

Figure 2. Trust Map

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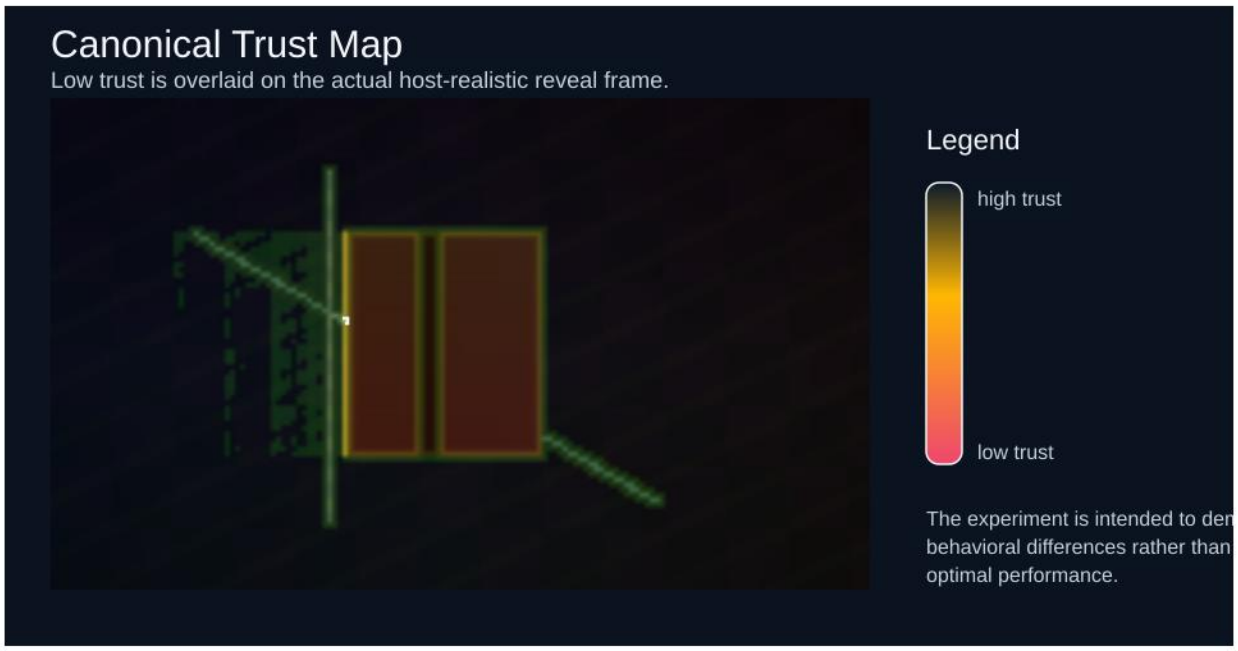

Low trust concentrates around the reveal event and motion-edge supervision region on the actual demo frame.

Figure 3. Before / After

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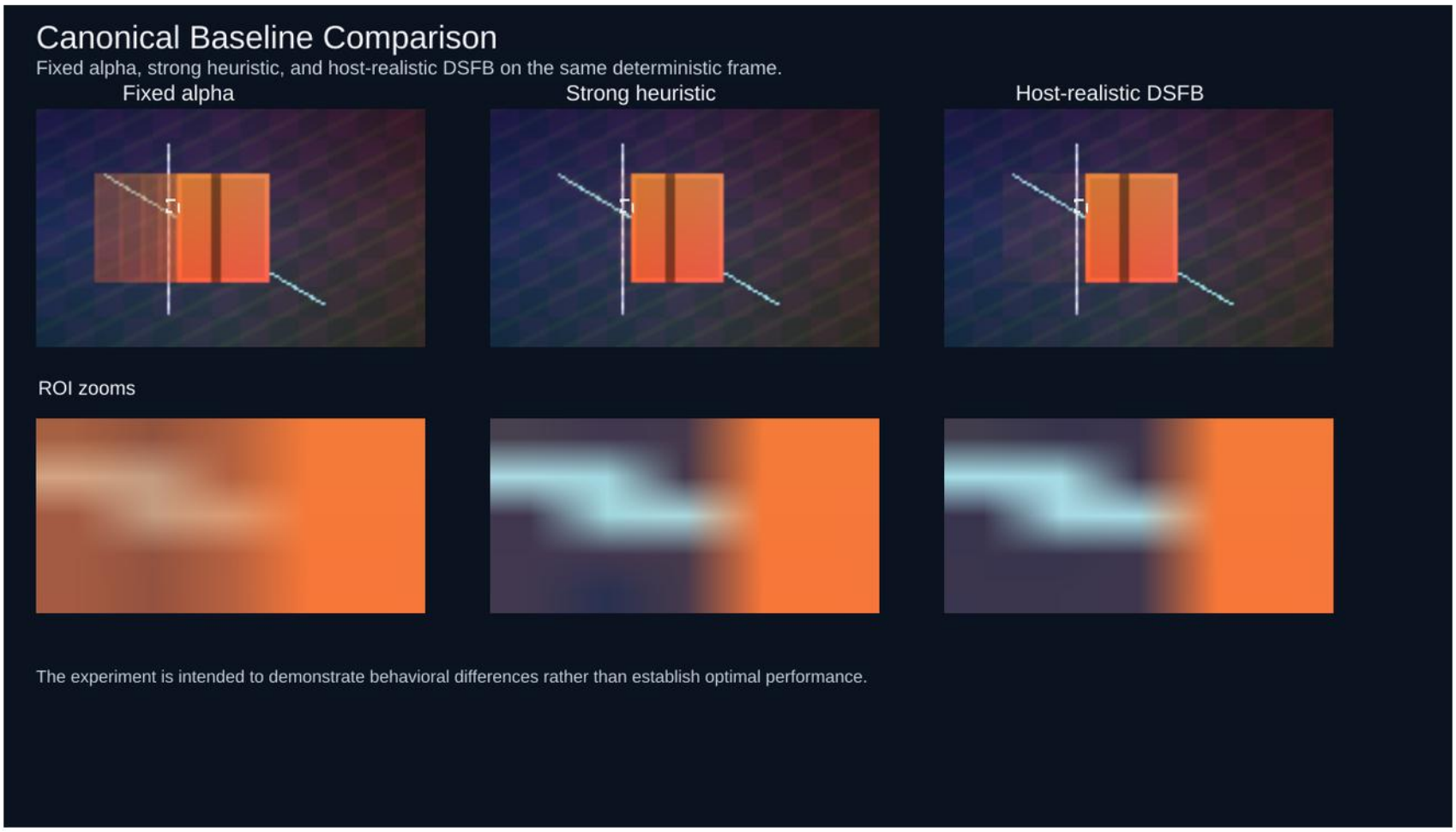

Fixed alpha, strong heuristic, and host-realistic DSFB on the same canonical comparison frame and persistence ROI.

Figure 4. Trust vs Error

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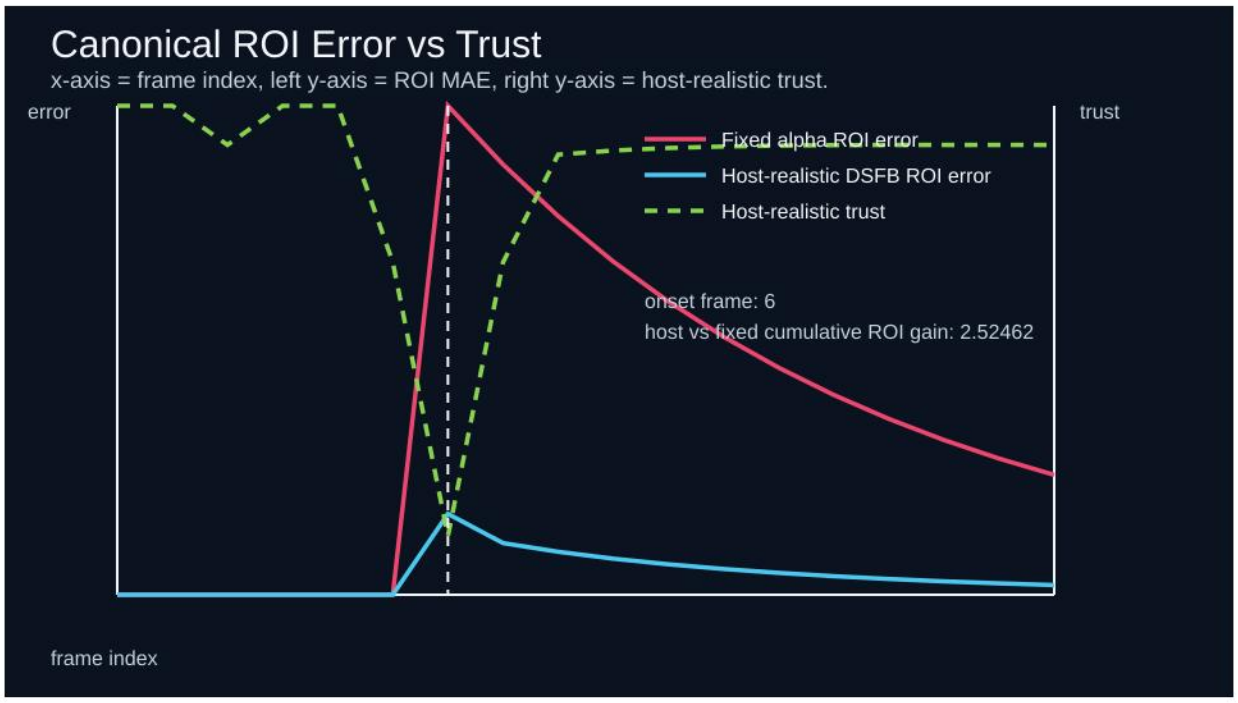

Frame index on the x-axis, ROI error on the left y-axis, and DSFB ROI trust on the right y-axis.

Figure 5. Intervention and Alpha

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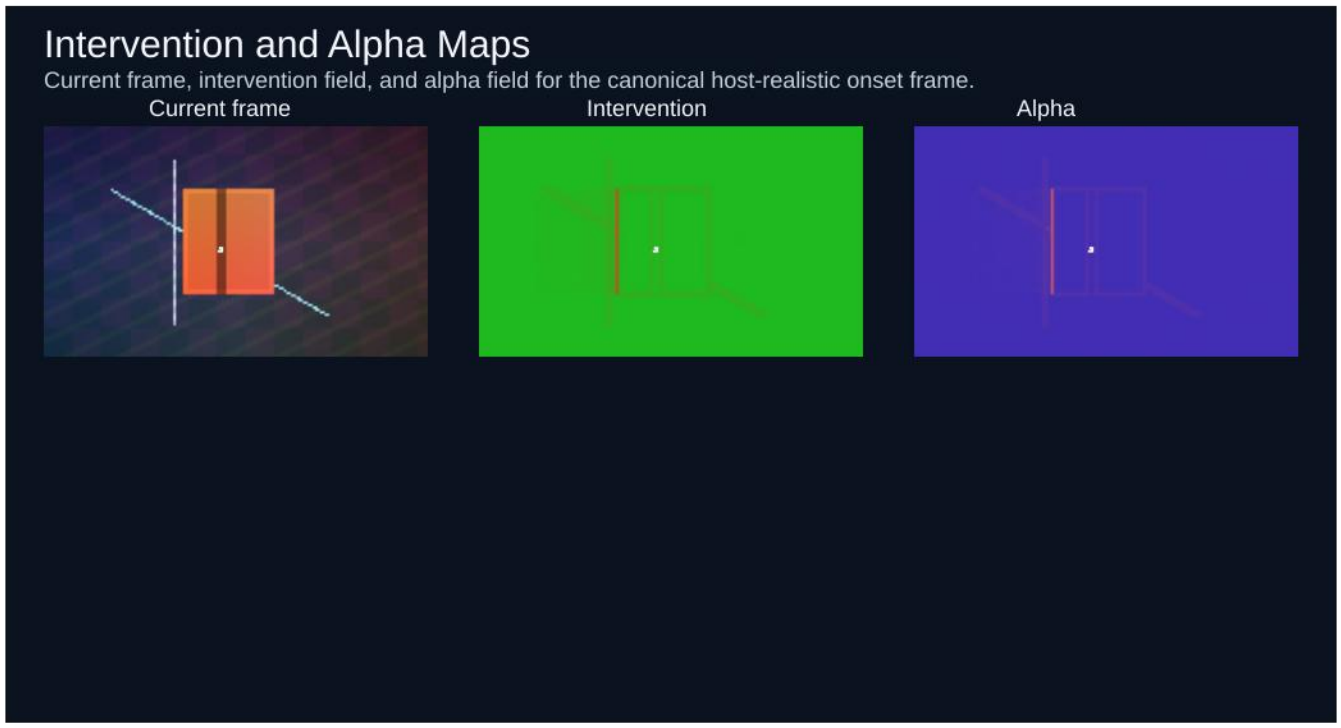

The intervention map and alpha field show where host-realistic DSFB actively suppresses stale history.

Figure 6. Aggregate Scenario View

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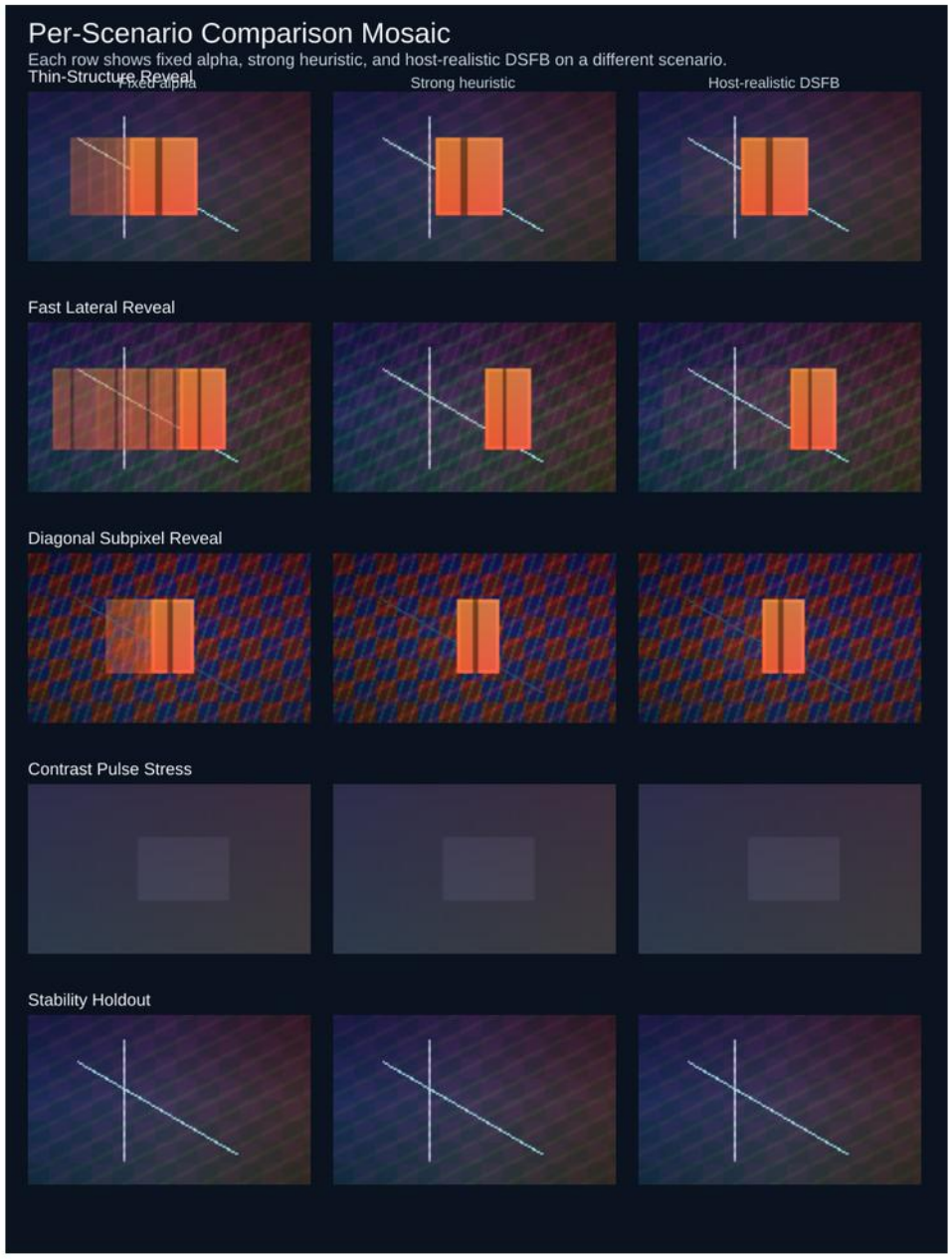
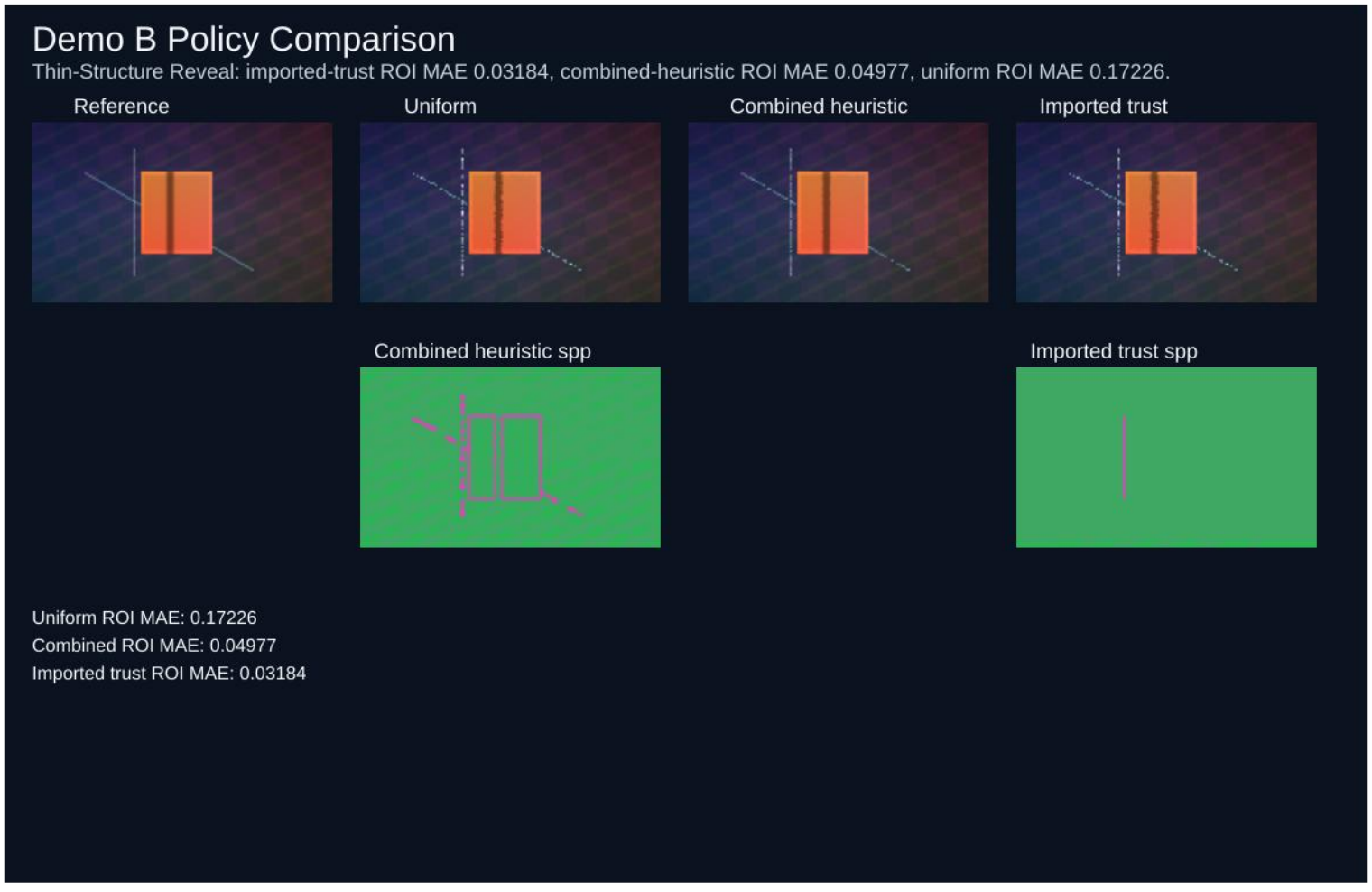


The expanded scenario suite makes the effect distribution and mixed outcomes visible instead of relying on one favorable case.

Demo B: Fixed-Budget Adaptive Sampling



At equal total sample budget, the imported-trust policy concentrates work in low-trust regions while remaining comparable to or better than cheap heuristic reallocators on the hardest cases. Uniform ROI MAE is 0.17226; combined-heuristic ROI MAE is 0.04977; imported-trust ROI MAE is 0.03184.

Demo B. Budget Efficiency

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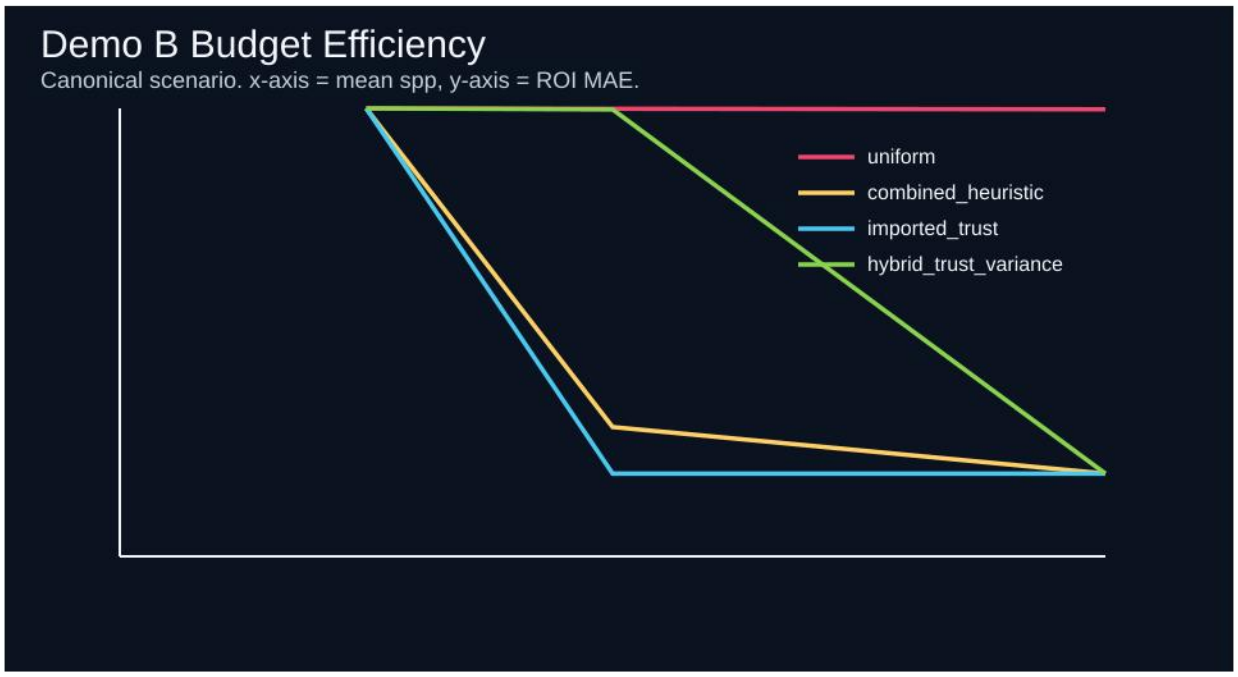

The budget-efficiency curve shows ROI MAE against mean spp for uniform, combined-heuristic, imported-trust, and hybrid trust-plus-variance allocation on the canonical sampling case.

Limitations and Scope

This remains a bounded synthetic artifact. It does not claim production-optimal TAA, measured GPU timing wins, field readiness, or universal superiority over commercial reconstruction systems.

PDF and ZIP export exist to make external review easier: a reviewer can archive the exact bundle from one run directory and inspect the same figures and metrics later without rerunning the notebook.

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